

# VOCAL COMMUNICATION IN FREE-RANGING AFRICAN WILD DOGS (LYCAON PICTUS)

by

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## Summary

From 1992-1998, a study of up to six radio-collared packs of free-ranging African wild dogs (*Lycaon pictus*) was conducted in Hwange National Park, Zimbabwe. Wild dogs were primarily observed during the day and data were collected across seasons on vocalizations and social context. An analysis of spectral properties, along with an examination of context and function, is presented. A vocal repertoire consisting of 11 vocal classes and 18 subclasses was identified.

The sound system of this socially obligate carnivore is among the most complex in Canidae. Several vocal classes, *i.e.* twitters, begging cries, and rumbles, appear to be unique. Heavy investment in high frequency sounds relative to other social canids is offset by a greater variety of low frequency sounds. More kinds of barks are found in its repertoire than have been reported for other canids. Sound mixing is an integral component of wild dog vocal communication and occurs via succession or superimposition of noisy and harmonic sounds. Mixing by succession occurs for all classes, while mixing by superimposition occurs for yelp/squeals, whines, moans, growls, and barks. The *Lycaon* vocal repertoire is highlighted

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by comparison with other social canids (*i.e.* wolves, bush dogs, dholes, and domestic dogs). Finally, attention is drawn to features in the wild dog sound system that are both consistent and inconsistent with the motivation-structural rules of Morton's model. While most sounds conform to the model, there appear to be anomalies between the expected design features of certain vocalizations (*i.e.* twitters, barks, and rumbles) and the contexts in which they are given. Some of these anomalies appear to reflect the highly cohesive social system of the wild dog.

## Introduction

The study of vocal communication in the African wild dog (*Lycaon pictus*) has lagged behind other scientific concerns in this socially complex carnivore. References to its vocalizations have tended to be either anecdotal, peripheral, or fragmented (*cf.* Kühme, 1965; Estes & Goddard, 1967; Schaller, 1972; Tembrock, 1976; Wilden *et al.*, 1998). A preliminary vocal ethogram for *Lycaon* was composed by Marten (1980). With the exception of the domestic dog (*Canis familiaris*), comprehensive vocal repertoires have been compiled for only three (*i.e.* *Canis latrans*, McCarley, 1975; Lehner, 1978a, b; *Vulpes vulpes*, Newton-Fisher *et al.*, 1993; *Canis lupus*, Schassburger, 1993) of the 34 species of Canidae (see Ginsberg & Macdonald, 1990).

Criteria for categorizing canid sounds have varied widely and have included causal antecedent, auditory quality, context, function, and spectral analysis (*e.g.* Cohen & Fox, 1976; Tembrock, 1976; Schassburger, 1993). Many vocalizations in this taxon tend to fuse in an acoustic continuum and are polymorphic. Existing nomenclature for classes upon which interspecific comparisons have been based consists largely of onomatopoeic transcriptions at times idiosyncratically applied. Despite such limitations, vocal taxonomy provides a useful window on larger patterns of behavior, particularly among social canids. For example, wild dog packs are typically composed of a breeding pair, non-reproductive helpers, and pups (Frame *et al.*, 1979). Acoustic communication accompanies most social interactions and appears to play an important role in intra- and interpack dynamics.

In this paper, the vocal repertoire of free-ranging wild dogs is presented, including mixed vocalizations. Sound mixing is a salient indicator of social complexity (Cohen & Fox, 1976; Schassburger, 1993) and an integral component of wild dog vocal communication. Major vocal classes are compared with those reported for other social canids, *i.e.* wolves, bush dogs,

dholes, and domestic dogs. Although pup vocalizations are mentioned, a detailed analysis will be presented elsewhere.

Features of the wild dog sound system both consistent and inconsistent with Morton's model of vocal communication (Morton, 1977, 1982) are also discussed. This model postulates motivation-structural (MS) rules to explain how natural selection may have shaped acoustic design features of vocalizations.

## Methods

### *Study area*

Located in the northwestern portion of Zimbabwe, Hwange National Park (HNP) encompasses 14 651 km<sup>2</sup>. Ninety two percent of the park is comprised of bushland and woodland (Rogers, 1993). The study area is located in the northern subregion and is *ca* 2200 km<sup>2</sup>, stretching from approximately 18°25' to 18°55' latitude and 26°00' to 26°45' longitude. The habitat consists of hills, kopjes, and flat stretches of basalt and Kalahari sands. *Colophospermum mopane* woodland and *Combretum* thicket are the dominant vegetation types (see Rogers, 1993).

### *Data collection and analysis*

Up to six radio-collared packs were followed across seasons from 1992-1998. Locations were recorded in Universal Transverse Mercator (UTM) coordinates using a Trimble Global Positioning System (GPS) or triangulation on 1:50 000 scale maps. Approximately 515 observation hours were made from a Land Rover 15-20 meters away from resting packs. Most recordings were made between dawn and dusk using either a Marantz PMD222 or Tascam DA-P1 audio recorder, and a K6 Sennheiser unidirectional microphone with a frequency response of 0.02-20.0 kHz. Approximately 20 hours of high quality recordings were made. A total of 321 spectrograms were produced across vocal classes using the program Spectraplus 4.0 and a narrow band filter. Most spectrograms ranged from 0.1-16.0 kHz. Spectrograms judged most representative of each vocal class were ultimately selected for analysis and presentation.

The principal method of vocal classification was spectral analysis, further interpreted by context, function and auditory quality, *i.e.* how they sounded to the unaided ear. Preliminary distinctions were drawn during observation periods based on auditory quality while simultaneously recording and noting the contexts in which sounds were given. Recordings were then spectrally analyzed. Measurements included fundamental, dominant frequency range, syllable length, and intersyllable and harmonic intervals. An attempt was made to avoid inflating or conflating classes. A comparison of spectrograms with similar measurements led to placing them in the same vocal class. Based on spectral analysis, it was thought more parsimonious to operationally lump yelps and squeals together even though they sounded differently to the unaided ear. Alternatively, whimpers and moans were placed in separate classes rather than identified as kinds of whines based on consistent spectral features indicative of discontinuity.

Refinement in spectral analysis led to the identification of various subclasses. Yelp barks were identified as a subclass rather than as a mixed vocalization based on stereotypy. The function of particular sounds, *i.e.* their placement into general behavioral categories (*cf.* Cohen & Fox, 1976), was determined by considering motivational affinities.

Existing terms for canid vocal classes were retained, as were those unique to the wild dog, *e.g.* 'twitter', 'begging cry' and 'hoo' (see Marten, 1980), while others like 'gurgle', 'buzz moan' and 'rumble' were introduced because no lexical equivalents could be found. 'Howl bark' was improvised with the caveat that it not be confused with the bark howl of other canids (*e.g.* Harrington & Mech, 1978). In the text, a 'syllable' denotes an unbroken trace of a spectrogram and an 'intersyllable interval', a temporal demarcation between the end of a syllable and the onset of another. 'Harmonic interval' refers to the frequency spacing generated by multiples of a fundamental. For brevity, 'dominant frequency range' *i.e.* the set of frequencies containing most acoustic energy, appears as 'frequency range'. Several syllables in close temporal proximity constitute a 'burst'. A 'bout' is an episode comprised of two or more bursts, and a 'volley' is a long distance acoustic exchange between conspecifics.

## Results

Although most vocal communication in the wild dog is encoded in acoustically graded signals, the sound system can be divided roughly into equal portions of low and high frequency. A total of 11 vocal classes and 18 subclasses were identified. Acoustic properties are presented in Table 1.

### *Description of vocalization types*

#### Twitters

Twitters are one of the most common and heterogeneous sounds (Figs 1a-1d). These primarily high-pitched, staccato-like emissions have fundamentals with middle range frequencies often suppressed. Four main subclasses have been identified: social, spar, mob, and attack (Marten, 1980).

Social — Social twitters are found in both low and high frequency ranges (Fig. 1a) with the latter predominant. Low frequency components are both noisy and harmonic. Although at times pronounced, these sounds are usually suppressed fundamentals of high frequency twitters. Both noisy and harmonic, high frequency components present a nearly vertical, but occasionally chevron-shaped appearance. Social twitters were used as vocal accompaniments in the greeting ceremony and, along with gestures, served to reinforce social bonds. Acoustic bursts lasted from 0.2-6.0 s and bouts up to 83.0 s.

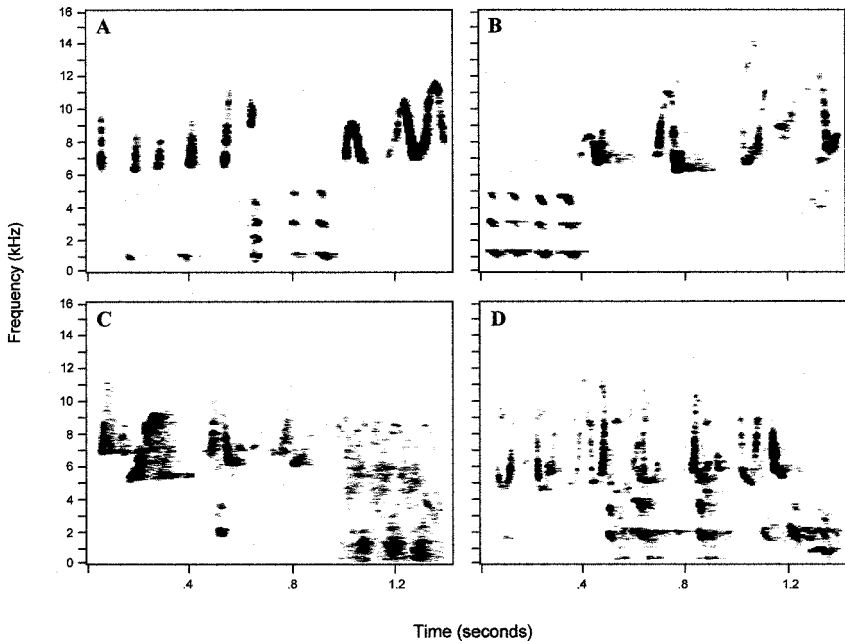


Fig. 1. Spectrograms representing acoustic structure of twitters: A. social; B. spar; C. mob; D. attack.

Social twitters were sometimes given during resting periods when individual pack members repositioned themselves or approached one another, and appeared to indicate a high level of arousal.

**Spar** — Low frequency components are more common in spar twitters (Fig. 1b). In contrast to social twitters, high frequency components are predominantly chevron or erratically shaped with spectral energy more unevenly distributed. Additionally, harmonic intervals in high frequency components tend to be greater and intersyllable intervals shorter than in social twitters. Noisy and harmonic components are found in both low and high frequency ranges. These twitters were given by subadults during vigorous play, and by adults in appeasement or when engaged in social investigation during pack formation (see McCreery, 1999). Acoustic bursts lasted from 0.3-1.0 s and bouts up to 6.3 s.

**Mob** — Low frequency components in mob twitters (Fig. 1c) are generally suppressed. Unlike social and spar twitters, high frequency twitters are mostly noisy with a breathy quality. They are often disyllabic with the first syllable more vertical, higher in frequency, shorter in length (Table 1),

TABLE 1. *Acoustic parameters of African wild dog vocalizations*

| Vocal class                     | Acoustic parameters <sup>1</sup>       |                          |                                |                                                   |                            |                          |
|---------------------------------|----------------------------------------|--------------------------|--------------------------------|---------------------------------------------------|----------------------------|--------------------------|
|                                 | Gross spectral type                    | Fundamental (kHz)        | Dominant frequency range (kHz) | Syllable length (s)                               | Intersyllable interval (s) | Harmonic interval (kHz)  |
| <i>Twitter</i>                  |                                        |                          |                                |                                                   |                            |                          |
| Social                          | L: noisy/harmonic<br>H: noisy/harmonic | L: 0.4-1.5<br>H: 0.4-2.5 | L: 0.4-5.0<br>H: 0.4-14.0      | L: 0.03-0.07<br>H: 0.02-0.04 (v)<br>0.06-0.1 (c)  | L: 0.1-0.3<br>H: 0.05-0.4  | L: 1.0-2.0<br>H: 0.1-0.3 |
| Spar                            | L: noisy/harmonic<br>H: noisy/harmonic | L: 0.4-1.5<br>H: 0.4-2.5 | L: 0.4-5.0<br>H: 0.4-14.0      | L: 0.03-0.07<br>H: 0.06-0.2 (c)<br>0.03-0.3 (e)   | L: 0.1-0.3<br>H: 0.05-0.2  | L: 1.0-2.0<br>H: 0.2-0.6 |
| Mob                             | M: noisy/harmonic<br>H: mostly noisy   | M: 1.7-2.0<br>H: 1.5-0.5 | M: 1.7-5.2<br>H: 1.5-14.0      | M: 0.02-0.05<br>H: 0.02-0.03 (v)<br>0.04-0.2 (e)  | M: 0.5-0.8<br>H: 0.05-0.5  | M: 1.5-1.8<br>H: 0.3-0.7 |
| Attack                          | L: mostly noisy<br>H: noisy/harmonic   | L: 0.2-2.0<br>H: 1.0-1.7 | L: 0.2-6.0<br>H: 1.0-12.0      | L: 0.03-0.06<br>H: 0.02-0.05 (v)<br>0.03-0.07 (u) | L: 0.1-0.4<br>H: 0.04-0.3  | L: 0.2-2.0<br>H: 0.2-0.6 |
| <i>Begging cry</i>              | harmonic                               | 0.5-2.5                  | 0.5-5.5                        | 0.03-2.2                                          | 0.02-0.6                   | 0.7-2.0                  |
| <i>Gurgle</i>                   | noisy/harmonic                         | 0.3-1.8                  | 0.3-5.5                        | 0.5-7.0                                           | 0.03-0.4                   | 0.7-0.9                  |
| <i>Yelp/Squeal</i> <sup>*</sup> |                                        |                          |                                |                                                   |                            |                          |
| Yelp                            | harmonic                               | 0.7-2.0                  | 0.7-7.0                        | 0.1-0.3                                           | N/A                        | 1.5-2.0                  |
| Whistle                         | harmonic                               | 0.7-2.0                  | 0.7-7.0                        | 0.4-1.0                                           | N/A                        | 1.5-2.0                  |
| Begging                         | harmonic                               | 1.0-2.0                  | 1.0-7.0                        | 0.5-1.0                                           | N/A                        | 1.5-2.0                  |
| <i>Whimper</i>                  | harmonic                               | 0.3-7.0                  | 0.3-8.0                        | 0.1-0.4                                           | 0.2-1.5                    | 0.5-2.0                  |
| <i>Whine</i>                    | harmonic/richly harmonic               | 0.4-2.0                  | 0.4-5.0                        | 0.2-2.7                                           | N/A                        | 0.4-1.0                  |

TABLE 1. (*Continued*)

| Vocal class                   | Acoustic parameters <sup>1</sup> |                   |                                |                     |                            |                         |
|-------------------------------|----------------------------------|-------------------|--------------------------------|---------------------|----------------------------|-------------------------|
|                               | Gross spectral type              | Fundamental (kHz) | Dominant frequency range (kHz) | Syllable length (s) | Intersyllable interval (s) | Harmonic interval (kHz) |
| <i>Moan</i>                   |                                  |                   |                                |                     |                            |                         |
| Full                          | richly harmonic                  | 0.1-0.6           | 0.1-1.2                        | 0.2-1.2             | N/A                        | 0.1-0.3                 |
| Buzz                          | richly harmonic                  | 0.2-0.4           | 0.2-3.0                        | 0.5-10.0            | N/A                        | 0.1-0.2                 |
| (Gurgle)                      | noisy/richly harmonic            | 0.2-1.8           | 0.2-3.0                        | 0.5-10.0            | N/A                        | 0.1-0.2                 |
| <i>Rumble</i>                 | mostly noisy                     | 0.2-0.3           | 0.2-0.5                        | 0.1-1.3             | 0.02-0.2                   | 0.2-0.5                 |
| <i>Growl</i>                  |                                  |                   |                                |                     |                            |                         |
| Social                        | mostly noisy                     | 0.2-0.5           | 0.2-1.5                        | 0.4-0.8             | N/A                        | 0.2-0.3                 |
| Alarm                         | mostly noisy                     | 0.3-0.5           | 0.3-2.0                        | 0.2-0.4             | N/A*                       | 0.2-1.5                 |
| <i>Bark</i>                   |                                  |                   |                                |                     |                            |                         |
| Alarm                         | mostly noisy                     | 0.2-0.5           | 0.2-4.5                        | 0.1-0.6             | N/A*                       | 0.2-1.5                 |
| (Common)                      | noisy/harmonic                   | 0.5-0.7           | 0.5-4.5                        | 0.6-0.7             | N/A*                       | 0.5-1.0                 |
| (Tremolo)                     | noisy/harmonic                   | 0.5-0.7           | 0.5-4.0                        | 0.2-0.3             | N/A                        | 0.3-1.0                 |
| Threat                        | mostly harmonic                  | 0.5-0.7           | 0.5-3.0                        | 0.7-1.2             | 4.1-9.3                    | 0.5-1.0                 |
| Howl                          | mostly harmonic                  | 0.3-0.5           | 0.3-4.0                        | 0.04-0.1            | 0.03-0.05                  | 0.3-1.0                 |
| Attack                        | mostly harmonic                  | 0.3-0.8           | 0.3-4.0                        | 0.1-0.4             | N/A                        | 0.5-1.0                 |
| Clear                         | mostly noisy                     | 0.6-1.7           | 0.6-6.5                        | 0.1-0.2             | N/A                        | 0.4-0.8                 |
| Yelp                          | harmonic                         | 0.6-0.8           | 0.6-0.8                        | 0.1-0.8             | 0.1-1.4                    | 0.8-1.5                 |
| <i>Hoo</i>                    |                                  |                   |                                |                     |                            |                         |
| <i>Pack call</i> <sup>2</sup> | noisy/harmonic                   | N/A               | N/A                            | N/A                 | N/A                        | N/A                     |

<sup>1</sup> Approximations. <sup>2</sup> Composite vocalization. \* See text. Key to abbreviations: L: low, M: medium, H: high frequency components; (v) vertical, (c) chevron, (e) erratic, (u) u-shaped.

and less amplified. Intervals between disyllables range from 0.05-0.2 s and 0.2-0.5 s between pairs. When present, middle frequency components are both noisy and harmonic and immediately precede the second syllable. Mob twitters were heard on eight occasions. They were given by adults while mobbing other pack members, pursuing prey, chasing hyenas and vultures from a kill, and by resting pups when approached by adults as a warning to lie elsewhere. Acoustic bursts in mob twitters lasted from 0.9-1.8 s and bouts up to 9.7 s. Although rarely heard, noisy low frequency, breathy components (see Fig. 1c) can be present with some harmonic energy concentrated between 0.6-1.5 kHz, and mostly noisy energy extending from 2.0-9.0 kHz. These sounds were issued in bursts lasting from 0.7-1.3 s with intersyllable intervals from 0.02-0.2 s.

**Attack** — In contrast to other twitters, lower components in attack twitters are more pronounced with lower fundamentals (Fig. 1d). Mostly noisy low frequency twitters that sound nasal are found around 1.5-2.0 kHz. Like social and spar twitters, attack twitters are both noisy and harmonic in high frequency ranges. High-pitched U-shaped twitters are diagnostic. These twitters were heard on six occasions. They were given by packs while chasing hyenas, during an interpack encounter, and by resting pups when approached by adults. Acoustic bursts lasted from 0.3-0.8 s and bouts up to 6.8 s.

### Begging cries

Begging cries (Figs 2a-2d) are spectrally heterogeneous emissions comprised of staccato (Fig. 2a) to long steady-state or modulated components (Fig. 2b). These high amplitude sounds often function as a form of food solicitation and, in the context of greeting ceremonies, may be viewed as a form of social solicitation (Kühme, 1965; Kingdon, 1977). They can also indicate distress and were given once by a dog while being mobbed by other pack members, and once by a subordinate female whose pup had just been cannibalized by the alpha female (Fig. 2c; see Robbins & McCreery, 2000). In the latter context, these sounds were spectrally noisier than any previously recorded. Acoustic bursts in begging cries lasted from 0.3-11.1 s, punctuated by periods of silence from 3.0-19.1 s. Bouts lasted from 0.5 to 137.0 s.

**Gurgles** — During peaks of arousal, begging cries can encapsulate noisy components (Fig. 2d) that range from 0.1-0.3 s in length and 1.0-3.0 kHz in



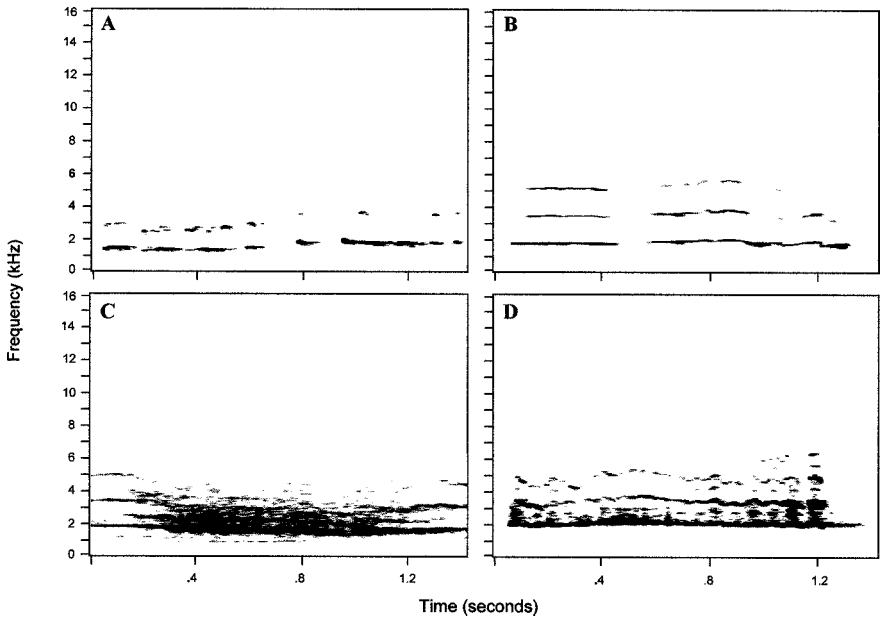


Fig. 2. Spectrograms representing acoustic structure of begging cries: A. staccato; B. steady-state and modulated; C. scream; D. gurgle.

frequency. Aside from buzz moans (see below), these cries have the longest syllable lengths of any vocalization in the repertoire (see Table 1).

### Yelp/Squeals

Yelp/squeals (Figs 3a-3c) are harmonic and high amplitude. Three subclasses were identified.

**Yelps** — Yelps (Fig. 3a) are short, sharp sounds that present a diagnostically convex spectral appearance. They were given in submission by subordinate dogs or pups either in anticipation of, or in response to, pain probably to inhibit potential injury. Yelps were also given in appeasement and solicitation during greeting ceremonies and begging cry displays, and were sometimes heard when packs chased hyenas.

**Whistle squeals** — Whistle squeals (Fig. 3b) can be either steady-state or modulated with syllable lengths over three times that of yelps. They were uttered in single or multiple syllables with intersyllable intervals from 0.04-0.06 s. These sounds were given in similar contexts as yelps.

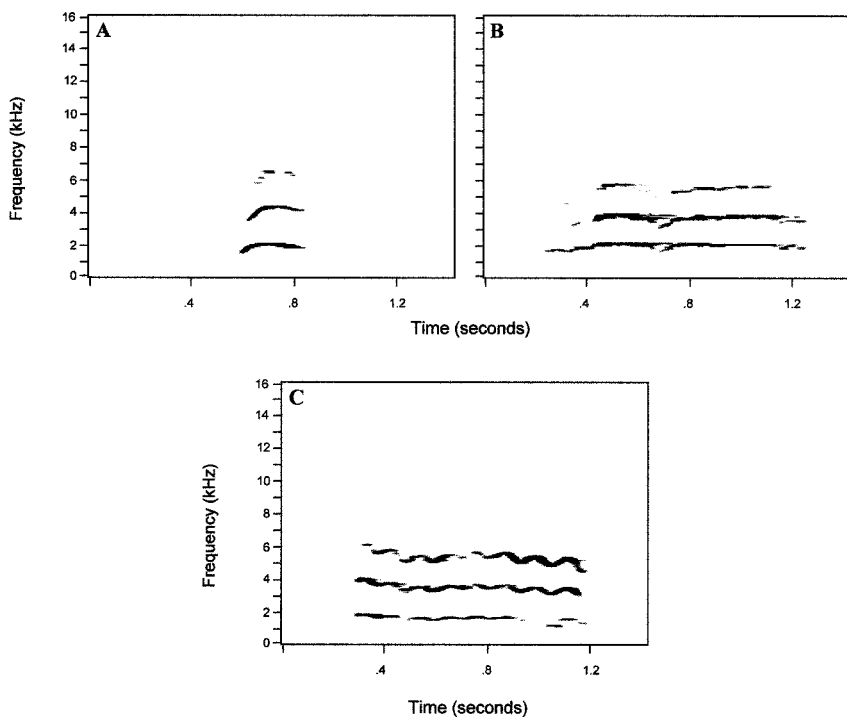


Fig. 3. Spectrograms representing acoustic structure of yelp/squeals: A. yelp; B. whistle squeal; C. begging squeal.

**Begging squeals** — Begging squeals (Fig. 3c) are diagnostically sinusoidal (wavelengths range from 0.05-0.1 s) with acoustic lengths comparable to whistle squeals. They have higher fundamentals and narrower frequency ranges than either yelps or whistle squeals (Table 1). These sounds were often given during greeting ceremonies and begging cry displays as expressions of solicitation as well as appeasement. Begging squeals were also heard accompanying spar twitters.

### Whimpers

Whimpers are low to high frequency sounds (Fig. 4) of low to medium amplitude. Spectral appearance varies from steady-state, modulated, and chevron-shaped, to disyllabic. In higher frequencies, multiple fundamentals are common with intervals from 0.6-0.8 kHz and middle ranges suppressed. Usually repetitive, whimpers were given when dogs approached one another or repositioned themselves during resting periods. They were also given

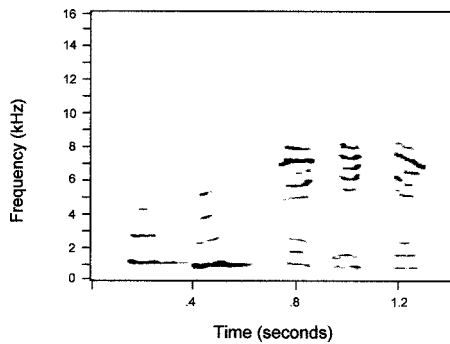


Fig. 4. Spectrogram representing acoustic structure of low and high frequency whimpers.

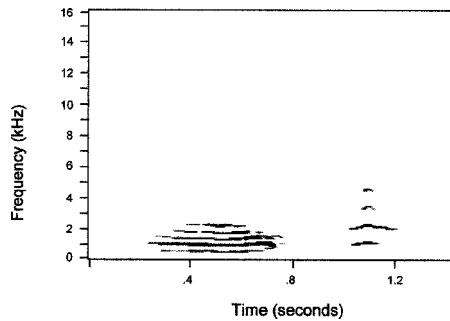


Fig. 5. Spectrogram representing acoustic structure of whines.

when adults called pups from the den, and during greeting ceremonies. On two occasions, low frequency whimpers were aired by dogs mobbing other pack members suggesting ambivalence.

### Whines

Whines (Fig. 5) are low to high frequency and low to medium amplitude. They can be steady-state or modulated. Unlike whimpers, harmonics are typically pronounced and syllables not repeated. They were heard given repetitively, however, in the rarely heard pack call (see below). Whines that sounded like meows and mews were issued by adults in greeting ceremonies. One of these ceremonies in a newly formed pack, consisted solely of these utterances. Although whines appear to be a form of solicitation or frustration, they were also aired on two occasions by dogs mobbing other pack members.

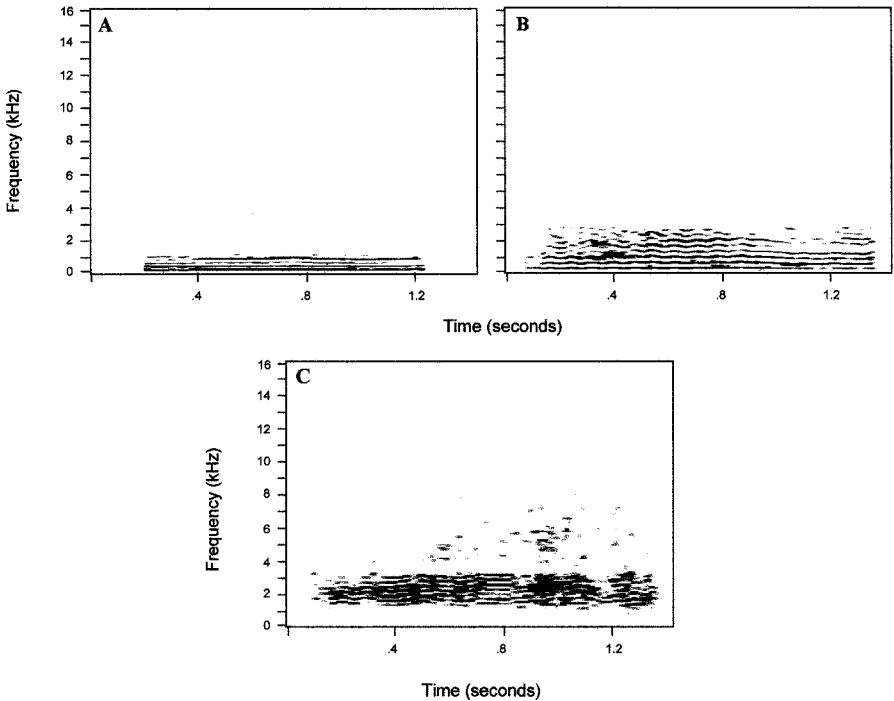


Fig. 6. Spectrograms representing acoustic structure of moans: A. full moan; B. buzz moan; C. buzz gurgle moan.

### Moans

Moans are richly harmonic, low to medium frequency and amplitude sounds (Figs 6a-6c) that can be either steady-state or modulated. In addition to the acoustic measurements already presented (Table 1), moans differ from whines in having a faster rate of frequency modulation. Two subclasses were identified.

**Full** — Full moans (Fig. 6a) have the shortest syllables and narrowest frequency range of the moans (Table 1). They appear to indicate frustration. These sounds were given by dogs trying to access kills and while digging at the opening of dens in which pups were resting.

**Buzz** — Buzz moans (Fig. 6b) have the longest syllables of any vocalization (Table 1). They are context specific utterances aired during the initial excitatory phase of high intensity greeting ceremonies and begging cry displays. Sometimes these sounds encapsulate noisy gurgles (Fig. 6c) that

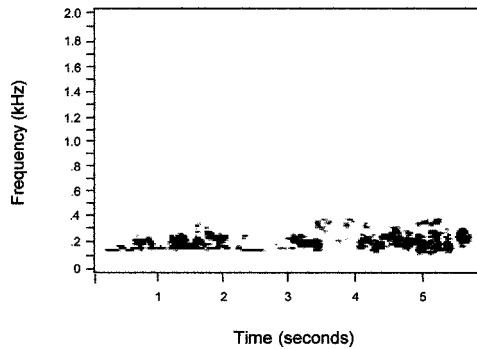


Fig. 7. Spectrogram representing acoustic structure of rumbles.

range from 1.0-2.0 kHz and last from 0.05-1.5 s. Buzz moans that include gurgles have greater fundamental frequency ranges.

### Rumbles

As assessed by frequency range, rumbles (Fig. 7) are the lowest sound in the repertoire (Table 1). Diagnostic are staccato-like guttural components with little time separation, strung sporadically to steady-state or modulated harmonic components. These low amplitude sounds were issued in acoustic bouts of about 10.0 s. Like buzz moans, rumbles are context specific and given during the initial excitatory phase of greeting ceremonies.

### Growls

Growls are low-pitched and mostly noisy (Fig. 8a). Two subclasses were identified.

**Social** — Social growls are low to medium amplitude with a relatively uniform distribution of energy over time and, when present, tightly banded harmonics (Table 1). They were heard during antagonistic interactions with other pack members.

**Alarm** — Alarms growls, in contrast to social growls, are shorter with greater frequency spread and tend to be more amplified, with most energy contained in the first half of the sound. These growls served to warn the pack when confronted with a possible threat and were sometimes given repetitively with intersyllable intervals from 0.5-6.3 s and bouts that lasted from 13.5-106.0 s.

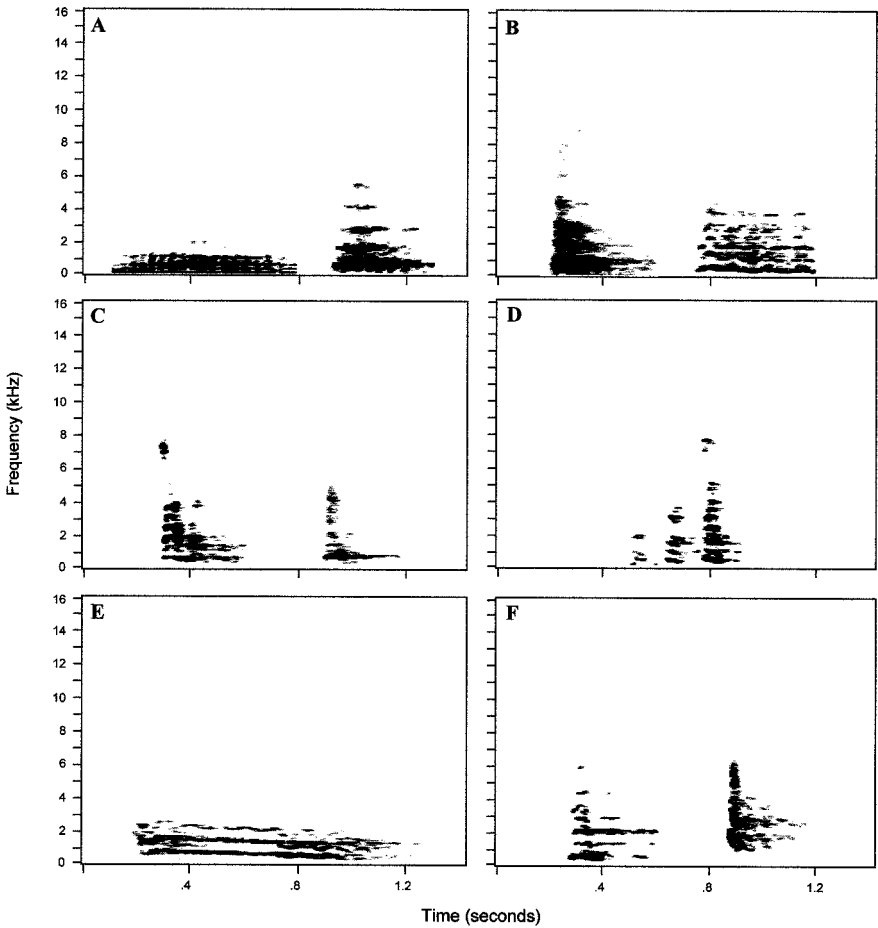


Fig. 8. Spectrograms representing acoustic structure of growls and barks: A. social and alarm growls; B. common and tremolo alarm barks; C. threat bark (disyllabic); D. attack barks; E. howl bark; F. clear and yelp barks.

## Barks

Barks (Figs 8b-8f) tend to have explosive onsets and are of greater frequency spread and amplitude than growls. Six subclasses were identified.

**Alarm** — There are two kinds of alarm barks (Fig. 8b): (1) common alarm barks are the most growl-like bark in spectral appearance and auditory quality. They express greater motivational intensity than alarm growls, but serve a similar function; (2) tremolo alarm barks indicate a higher level of arousal than common alarm barks. They are also more harmonic and longer

in duration, with a more uniform distribution of energy over time. These high intensity barks were given when lions were detected or when humans approached on foot. Alarm barks were occasionally issued repetitively with intersyllable intervals and bout lengths similar to alarm growls (see above).

**Threat** — Threat barks tend to be shorter than alarm barks and disyllabic (Fig. 8c). Each syllable is the same relative length, separated by intervals ranging from 0.1-0.3 s. The first part of the first syllable is often stressed and harmonic, presenting a convex spectral appearance. These barks may express motivational ambivalence. They were given toward pups soliciting food and appeared to signal muted aggression.

**Attack** — Attack barks are the shortest bark and one of the most harmonic (Fig. 8d). Harmonics tend to be strong and tightly banded. Of low to medium amplitude, these sounds can be entrained in a series of increasing syllable length and amplitude. To the unaided ear, they have a nasal quality. Attack barks were issued by pack members chasing hyenas.

**Howl** — Howl barks (Fig. 8e) are the longest barks and have the least frequency spread. Uttered with a tremolo, these mostly harmonic sounds were heard dropping 0.3-0.5 kHz in frequency. They were given by individual dogs while running excitedly during an interpack encounter.

Unlike other barks, clear and yelp barks (Fig 8f) span the motivational continuum (*sensu* Morton, 1977, 1982). They are given when highly aroused.

**Clear** — Clear barks are mostly harmonic with energy concentrated in the fundamental and from 2.0-2.5 kHz. Although wild dogs do not often emit these clear, sharp barks commonly associated with domestic dogs, they were sometimes given by adults in alarm, and, on two occasions, when adults interacted affiliatively with pups.

**Yelp** — Yelp barks are mostly noisy. Compared to other barks, they have the highest fundamentals and greatest frequency ranges (Table 1). Aggressive yelp barks were given when packs mobbed hyenas. Once, an adult issued an affiliative yelp bark toward pups at the den.

## Hoo

Hoos are pulsed, usually monotonic, concatenations of single syllables (Fig. 9). They are the only vocalization in which the fundamental and

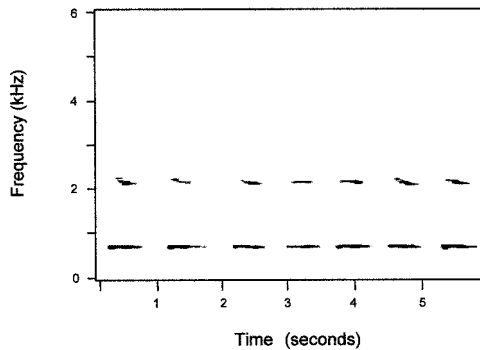


Fig. 9. Spectrogram representing acoustic structure of hoos.

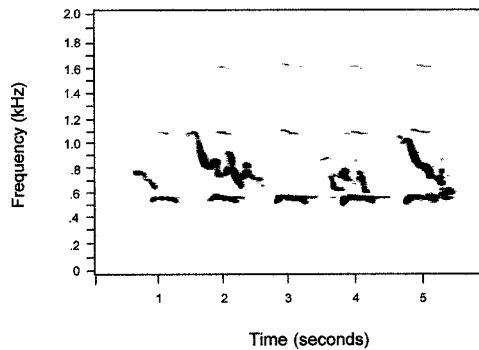


Fig. 10. Spectrogram representing acoustic structure of hoos and whines in a pack call.

frequency ranges coincide. As assessed by context, hoos are the only known long distance contact call. These sounds were heard most frequently when dogs became separated while hunting and can indicate distress. Once a dog at the den waiting the return of the pack from the hunt was heard hoo calling. Hoo call bouts lasted from 2.0-17.0 s.

#### Pack call

Pack calls are composites given within the context of interpack encounters. They can include hoos, whines, whimpers, yelp/squeals, growls, barks, and twitters. Bouts lasted from 36.2-62.4 s. These calls were chorused by both packs simultaneously or issued in volleys. They were given once outside the context of an interpack encounter. Periodically interwoven with whines (Fig. 10), they were aired when pack members reunited. It is possible that initially the two groups mistook each other for unknown conspecifics.



### *Mixed vocalizations*

Mixing of vocal classes occurs in the wild dog sound system from low to high frequency via succession or superimposition (*i.e.* without a temporal gap). Successive mixing occurs across all classes and between noisy and harmonic sound types. Mixing by superimposition was heard for five sound classes,

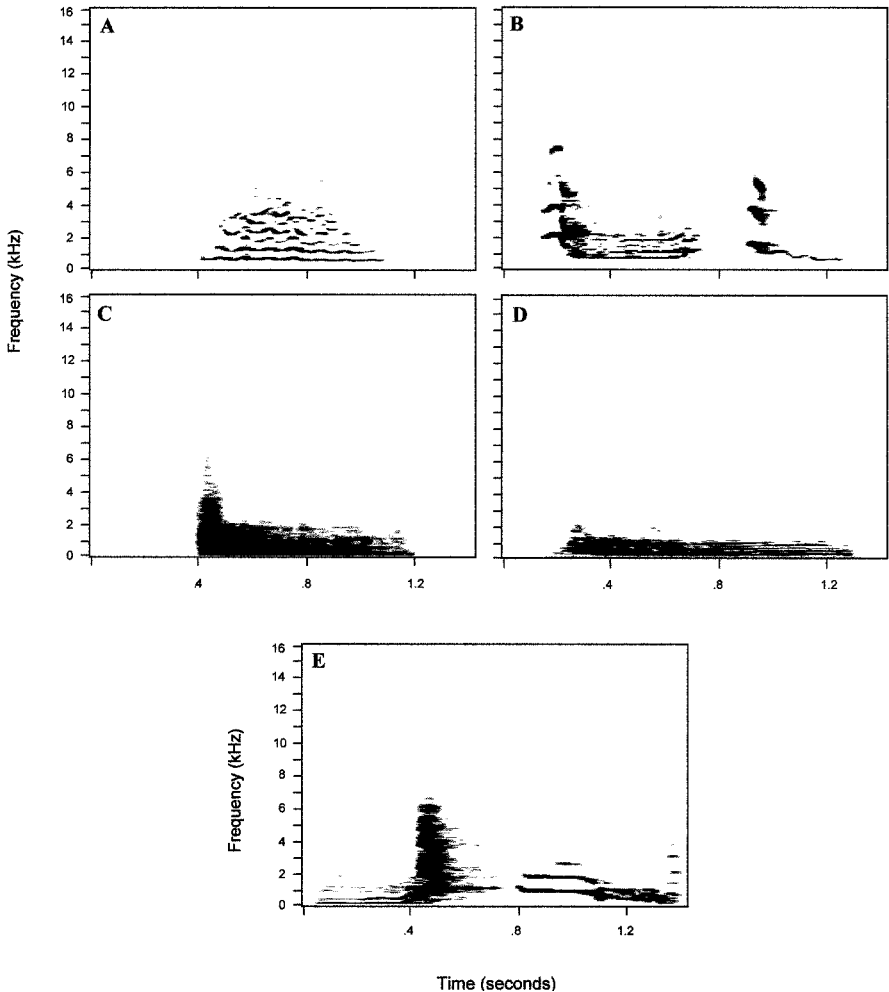


Fig. 11. Spectrograms representing acoustic structure of some superimposed mixed sounds. Harmonic with harmonic: A. whine-begging squeal; B. yelp-whine and yelp-begging squeal. Noisy with noisy: C. alarm bark-alarm growl. Noisy with harmonic: D. social growl-moan; E. moan-yelp bark and whine-social growl.

*i.e.* yelp/squeals, whines, moans, growls, and barks. Examples of mixing by superimposition are given in Figs 11a-11e. Mixing of harmonic sound types is displayed in the whine-begging squeal (Fig. 11a), yelp-whine and yelp-begging squeal (Fig. 11b). Mixing of noisy sound types is represented by the alarm bark-alarm growl (Fig. 11c). Mixing across noisy and harmonic sound types was rarely heard, but occurred in the social growl-moan (Fig. 11d), and the moan-yelp bark and whine-social growl (Fig. 11e).

## Discussion

### *Vocal taxonomy and comparison with other social canids*

The African wild dog has been classified among the most social of all canids (Fox, 1975). Social complexity is expressed in a rich array of vocalizations, some of which appear to be unique (see below). Semantic import is nested in the larger social domain (*cf.* Moran, 1987). Depending upon the motivational state and level of arousal of pack members, sounds issued by individuals can give rise to cascading group effects. For example, the onset of the greeting ceremony is frequently initiated by a single dog running up to another with head shoulder-height, mouth in a gape, and ears folded back, while initiating muzzle-to-muzzle contact. High intensity displays involve ritualized parallel walking accompanied by low amplitude rumbles and buzz gurgle moans that, with increasing motivational intensity, are replaced by twitters, begging cries, yelp/squeals, whines, and whimpers. Greeting behavior in adults may have developed from infantile begging (Kühme, 1965). Muzzle-to-muzzle contact, including lip licking and biting, appears to be a 'symbolic solicitation for food' (Kingdon, 1977).

Most wild dog vocalizations do not engage conspicuous oro-facial musculature or diaphragm. This contributes to a kind of 'ventriloquism' heightened by a dark face that often masks expression (J. Malcolm, pers. comm.; pers. obs.). Twittering occurs, however, with slightly retracted lips and quivering jaws (Schaller, 1972; pers. obs.). The dhole (*Cuon alpinus*) may be the only other canid to produce similar bird-like sounds (Johnsingh, 1982; Fox, 1984), but to date no spectrograms have been made. While wild dog social twitters are exclusively affiliative, spar twitters can have an antagonistic component as play escalates (Table 2). High-pitched mob twitters are antagonistic. The low-pitched sounds that sometimes accompany mobbing bouts appear to be

primarily 'unvoiced' (*sensu* Tembrock, 1963; Cohen & Fox, 1976) and produced by panting. In close contact antagonistic encounters, the depth and rate of respiration may correspond with the level of arousal (*cf.* Cohen & Fox, 1976). Attack twitters are high-pitched and 'chatterly' (Marten, 1980) with lower components that sound 'nasal' (Marten, 1980). They are issued in similar contexts as mob twitters.

Begging cries appear to be unique to the wild dog (Table 2) and can be aired by any member in the pack particularly during the greeting ceremony. These sounds are most pronounced around the time of denning when issued by the whelping female as part of a high intensity display. Approaching other pack members, they are emitted with a lowered head and gaping mouth while initiating muzzle-to-muzzle contact in an effort to induce regurgitation. The onset of the begging cry display commonly includes buzz moans that transition as excitement mounts into begging cry gurgles, mixed with twitters and yelp/squeals.

Yelp/squeals play an active role in adult acoustic displays. Although these sounds can serve different functions in wild dogs, wolves, and domestic dogs, they are commonly given in submission and distress (Table 2). Aside from wild dogs, begging squeals have been reported only in wolf neonates (Coscia *et al.*, 1991).

Given the daily activity pattern of wild dogs (see McCreery, 2000), repetitive whimpers are one of the most common affiliative vocalizations. Similar to other social canids, these sounds are used in greeting and can express submission/appeasement, distress and the need for social contact (Table 2). When lower frequencies are suppressed, they resemble squeaks reported in wolves (Field, 1975), but are considerably higher in frequency.

Whines serve a similar function as whimpers (Table 2). Some wild dog whines are modulated. In domestic dogs, modulated whines are formed when the tongue alternately opens and blocks air passages within the vocal cavity (Crisler, 1958, cited by Cohen & Fox, 1976). Wild dogs may form these sounds in the same way. To the unaided ear, some of their whines sound catlike. With the exception of vulpine species, meows and mews have been reported only in canine neonates (Cohen & Fox, 1976).

Full moans are low to medium amplitude emissions uttered by wild dogs when frustrated or distressed. They are issued by wolves in ambivalence and play as well (Schassburger, 1993). 'Groans' given by distressed domestic dogs (J.P. Scott & B. Roscoe, cited by Bleicher, 1963), spectrally resemble

TABLE 2. *Vocal classes and function in the African wild dog and other social canids*

| Function*                       | Vocal class     |             |                     |                                                               |            |      |        |                       |                |            |
|---------------------------------|-----------------|-------------|---------------------|---------------------------------------------------------------|------------|------|--------|-----------------------|----------------|------------|
|                                 | Twitter         | Begging cry | Yelp/Squeal         | Whimper                                                       | Whine      | Moan | Rumble | Growl†                | Bark           | Hoo        |
| Greeting                        | WD <sup>1</sup> | WD          | WD,D <sup>2,3</sup> | WD,W <sup>4,5,6</sup> ,<br>B <sup>7,H</sup> <sup>8,9</sup> ,D | WD,W,B,H,D | WD   | WD     | W,D                   | D              | -          |
| Play                            | WD              | -           | D                   | D                                                             | D,W        | W    | -      | W                     | W,D            | -          |
| Submission/Appeasement          | WD              | -           | WD,W,D              | WD,W,D                                                        | WD,W,D     | -    | -      | -                     | -              | -          |
| Defense/Alarm                   | -               | -           | D                   | D                                                             | D          | W    | -      | WD,W,<br>B,H,D        | WD,W,<br>B,D,H | -          |
| Threat/Attack                   | WD              | -           | -                   | -                                                             | -          | W    | -      | W,B<br>WD,W,<br>B,H,D | WD,W,<br>B,H,D | -          |
| Distress                        | -               | WD          | WD,W,D              | WD,W,B,D                                                      | WD,W,B,D   | WD,W | -      | W                     | D              | WD         |
| Care- or contact-seeking        | WD              | WD          | WD,D                | WD,W,B,H,D                                                    | WD,W,B,H,D | WD   | -      | -                     | D              | -          |
| Contact seeking<br>(long range) | -               | -           | -                   | -                                                             | -          | -    | -      | -                     | D              | WD,<br>B,H |

Key to abbreviations: WD wild dog. W wolf. B bush dog. H dhole. D domestic dog. <sup>1</sup>Marten, 1980; this study; <sup>2</sup>Bleicher, 1963; <sup>3,4</sup>Cohen & Fox, 1976; <sup>5</sup>Harrington & Mech, 1978; <sup>6</sup>Schassburger, 1993; <sup>7</sup>Brady, 1980; <sup>8</sup>Johnsingh, 1982; <sup>9</sup>Fox, 1984. \* Adapted from Cohen & Fox, 1976.

† Includes snarls in wolves and domestic dogs.

moans. Richly harmonic buzz moans and mostly noisy rumbles are contact seeking, transitional sounds given during the initial excitatory phase of greeting ceremonies. Rumbles are unlike any vocalization previously reported in canids.

Growls and barks have been reported in all social canids (Table 2). They can vary in tonality, amplitude, and onset depending on context. Wild dog social growls are limited to conflict within the pack and, unlike in most other canids, are rare. This finding is not unexpected given the high degree of cooperation that typifies *Lycaon* social organization. Wild dogs may issue play growls as well, but the sample size in this study was too small for classification.

There are more subclasses found in barks than in any other vocalization. Indeed, there are more kinds of barks in the wild dog repertoire than have been reported for any other canid. Why this is the case is unclear, since it seldom barks. In the wild dog sound system, growls and barks are primarily alarm signals. High intensity alarm barks have a pronounced tremolo, with each emission commonly accompanied by a nod (*sensu* Frame, 1985) while looking in the direction of the threat. Dogs have been observed (S. Childes, pers. comm; pers. obs.) emitting these barks while rearing back on hindlegs to better assess the presence of lions. Threat barks were given by adults toward pups in the context of pups begging for food. K. Marten (pers. comm.) reported subordinates issuing them defensively suggesting possible retaliation.

Some vocalizations like yelp and clear barks express similar high levels of arousal, but lack motivational specificity. Aside from the contexts in which they were heard in this study (see: *Results*) Kühme (1965) reported that they were given affiliatively by pups when meeting adults returning from the hunt. In contrast, Schassburger (1993) concluded that growls and barks given by wolves are always antagonistic. This latter observation conflicts with Cohen & Fox (1976), however, who found that wolves growl when greeting. Domestic dogs sometimes bark in this context as well (Cohen & Fox, 1976).

Unlike some canids (see Theberge & Falls, 1967; Cohen & Fox, 1976; Fox, 1984), wild dogs and bush dogs (*Speothos venaticus*, Brady, 1981) do not howl (Table 2). The high amplitude, medium range howl bark comes closest to it, but falls on the short end of howling in syllable length when compared to wolves (Theberge & Falls, 1967). Heard only in the context of

an interpack encounter, this sound may function as a high intensity alarm and/or threat call.

Long distance contact calls are common in Canidae and assume the form of either pulsed trains (Brady, 1981; Johnsingh, 1982; Fox, 1984; Durbin, 1998), or long acoustic streams (Theberge & Falls, 1967). Wild dogs are primarily crepuscular carnivores. In habitats like HNP where thick bush often makes visual detection difficult, the role of vocal behavior is greatly magnified. When filtering through bush in search of prey, they frequently become separated and may hoo call. Hoo calling occurs with the head held slightly lower than the shoulders and a gaping mouth that oscillates slightly with each emission. In contrast, dholes (Durbin, 1998) whistle and wolves (Theberge & Falls, 1967) howl by lifting their muzzles. While wild dogs hoo either standing or running, wolves howl only when standing still.

Although the height of the hoo call from the ground and its frequency are too low to be optimal for long distance communication in thick bush (Marten, 1980), pulsed and generally monotonic design features contribute to detectability when assessed against ambient noise levels. Hooing is issued in bouts or volleys of varying length and cadence, and localized by binaural phase difference using prominent parabolic ears. These sounds appear to play a role in emigration when opposite-sex groups are trying to locate one another (Frame *et al.*, 1979; pers. obs.).

Some wild dog vocalizations like pack calls are rare and recording them is largely fortuitous. Pack calls are cacophonies that can include most vocal classes. Unlike wolf pack calls (Klinghammer & Laidlaw, 1975; Harrington & Mech, 1978), wild dog pack calls are not given during the greeting ceremony nor are they associated with the breeding season. With one exception (see: *Results*), these calls were heard only during interpack encounters.

In wolves, interpack encounters have resulted in fighting and death (Mech, 1977). One of the functions of wolf pack calls appears to be as a spacing mechanism to avoid encounters (Harrington & Mech, 1979; Harrington, 1987). In contrast, wild dog pack calls (as assessed by radio telemetry) are issued when packs meet and shortly after separation. While clashes have been reported, they rarely result in death (Creel & Creel, 1994). Wild dog packs seem to prefer to avoid one another, with a larger pack usually chasing off a smaller one (Creel & Creel, 1994). It is possible that some high frequency sounds express motivational ambivalence toward another

pack, while simultaneously reaffirming intrapack bonds. Low frequency, high amplitude sounds like pack hoos may serve a dual purpose as well: to assess a pack's relative location, and to reassemble members after the initial excitement and scattering of an interpack contact has been made.

### *Mixed vocalizations*

Sound mixing is a salient measure of social complexity and reportedly more common in social than solitary canids (Cohen & Fox, 1976). Sounds bridging the gap between vocal classes either by succession or superimposition greatly enhance the communicative value of a sound system reflecting different, occasionally conflicting, motivations, thus serving to modulate behavior (Cohen & Fox, 1976; Schassburger, 1993). In highly complex social systems, we would expect to find a spectrum of sound qualities from noisy to harmonic with many points along a motivational gradient (*cf.* Morton, 1977). In wild dogs, data from this study indicate that successive mixing occurs for all classes, while mixing by superimposition occurs for half. In wolves, mixing by superimposition occurs for all major sound classes (Schassburger, 1993). With the exception of whimpers in wild dogs, mixing by superimposition occurs for all vocal classes that wild dogs and wolves have in common. Sound mixing across noisy and harmonic classes rarely occurs in the wild dog, wolf (Schassburger, 1993), and domestic dog (Cohen & Fox, 1976). In wolves (Schassburger, 1993) and wild dogs, most mixing involves two classes, with greater mixing in the low frequency noisy range than in the high frequency harmonic. In wolves, harmonic sounds tend to be more stereotyped and appear to be biologically designed to convey relatively unambiguous information, in contrast to noisy ones that are more elaborate and permit a wider range of motivational fluctuation (Schassburger, 1993). Results from this study indicate that this is applicable to wild dog vocalizations as well.

### *Wild dog vocalizations and Morton's model*

The relationship of acoustic design to motivation is an intriguing one. Morton's motivation-structural (MS) model (which primarily applies to short-range communication) postulates two physical dimensions to vocalizations: frequency and sound quality (Morton, 1977, 1982; Owings & Morton, 1998). Morton hypothesized that reciprocity generally obtains between frequency

and tonality such that the higher the frequency, the more harmonic the sound and, conversely, the lower the frequency, the noisier the sound. High-pitched, harmonic sounds should be positively correlated with fear and affiliative behavior at one end of the continuum and low-pitched, noisy sounds with aggression at the other (but see August & Anderson, 1987; Hauser, 1993). This bipartite model is reportedly consistent with the sound systems of bush dogs (Brady, 1981) and timber wolves (Schassburger, 1990). In the case of the wild dog sound system, however, there appear to be anomalies between the expected design features of certain vocalizations (*i.e.* twitters, barks, and rumbles) and the contexts in which they are given.

Twitters can be viewed along the (MS) continuum with the social twitter at one end and the attack twitter at the other. Social and spar twitters are generally high-pitched with both noisy and harmonic components. Following the (MS) model, however, one would not expect the presence of noisy components, particularly in higher frequency ranges, in the exclusively affiliative social twitter. Noisy components in the spar twitter may be more expected as play escalates. While the mostly noisy mob twitter agrees in sound quality with the (MS) model, it is considerably higher in pitch than would be expected with lower components generally suppressed. Although, at times, this twitter may express ambivalence (*cf.* McCreery, 1999), it is used in some of the same contexts as the attack twitter (Marten, 1980; see: *Results*). Additionally, some of the design features of the attack twitter are not altogether consistent with the (MS) model as well. Whereas lower components are noisier and more pronounced than for other kinds of twitters, higher ones appear to be more harmonic than those of the mob twitter.

The (MS) model proposes that barks, which tend to be in the noisy portion of the sound system, vary in frequency and sound quality depending upon motivation, with most being somewhere in the middle of the continuum. The noisy common alarm bark, which one would expect based on spectral features to be at the aggressive end of the continuum, is often the precursor to flight. In contrast, the high intensity tremolo alarm bark is more consistent with the model in that its harmonic structure probably represents fear. Even though in the threat bark there is the potential for at least muted aggression, motivational ambivalence appears to be amplified in the mostly harmonic first portion of the bark. At the antagonistic end of the (MS)



continuum, the short-range attack bark is unexpectedly one of the most harmonic.

Typically issued in long acoustic trains, close contact rumbles are low frequency and amplitude emissions that belong primarily to the noisy sound type. While the absence of tonality in low frequencies is consistent with the model, this vocalization was heard only during greeting ceremonies.

The anomalies discussed here with respect to twitters and barks may be explained in terms of the generally muted nature of aggression characteristic of wild dog social organization. In *Lycaon*, sociality is taken to an extreme (McCreery, 1999). Selection appears to have favored a highly cohesive social system in which affiliative and submissive gestures predominate (Kühme, 1965; Malcolm, 1979). While the wild dog sound system can be roughly divided into equal portions of low and high frequency, the high-pitched, distance-reducing sounds of puphood appear to be primarily used (see Marten, 1980; pers. obs.). According to the (MS) model (Morton, 1977, 1982), heavy investment in high frequency harmonic sounds is not unexpected given the gregarious nature of this species (Marten, 1980). Relative to other social canids, the low investment afforded long distance communication exemplifies *Lycaon* social organization as well (Marten, 1980).

## Conclusion

The vocal repertoire of the African wild dog is one of the most complex in Canidae. An analysis of vocal classes with respect to spectral features, context, and function has been presented as well as a comparison with other social canids. Mixed sounds have been considered for the first time in this species. It is probable that further research will reveal more examples of mixing than have been presented here. Aside from some vocalizations not previously reported for other members of this family, particularly remarkable is the heavy investment in high frequency emissions. A disproportionately high investment in these sounds appears to be counterbalanced by a greater variety of low frequency sounds. Attention has been drawn to features that are both consistent and inconsistent with Morton's model. A more complete understanding of wild dog vocal behavior will emerge only when the ontogeny of sound classes and a fuller description of accompanying gestures

are provided alongside those features already considered. This approach will be especially useful in still much needed comparative studies of social canids.

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